B. Hyperparameters

Here, we list the hyperparameters used in the final experiments in Section 6.

Table 6 | **Hyperparameters for The AI Scientist.**

Category	Hyperparameter	Value
Idea Generation	Number of Idea Reflections	3
	Number of Novelty Search Rounds (Semantic Scholar)	10
Experiment Execution	Max Experiments	5
	Max Experiment Attempts	4
	Experiment Timeout	7200 seconds
	Plotting Timeout	600 seconds
Paper Writing	Number of Citation Rounds (Semantic Scholar)	20
	Number of LaTeX Error Correction Rounds	5
Reviewer	Number of Reflections	5
	Number of Fewshot Examples	1
	Number of Ensembled Reviews	5
	LLM Temperature	0.1

C. Progression of Generated Ideas

We visualize the progression of ideas across a run of THE AI SCIENTIST on the "Grokking" template described in Section 6.3 using Sonnet 3.5. The first idea is the seed idea, all subsequent ideas are AI-generated.

Seed Idea - batch_size_grokking

```
"Name": "batch_size_grokking",
"Title": "Batch Size Grokking: Assessing the impact of the training
batchsize on the grokking phenomenon",
"Experiment": "Modify the experiments to dynamically adjust the batch size
during training, starting with a small batch size and gradually increasing
it. This could potentially lead to faster generalization on the validation
set.",
"Interestingness": 6,
"Feasibility": 4,
"Novelty": 4,
"novel": true
```

Idea 1/50 - model_size_grokking

```
"Name": "model_size_grokking",
"Title": "Investigating the Impact of Model Size on the Grokking
Phenomenon",
"Experiment": "Modify the Transformer class to accept variable number of
layers and dimension sizes. Test models with 1, 2, 4, and 8 layers, and
dimension sizes of 64, 128, 256, and 512. For each dataset and model size,
track the step at which grokking occurs (defined as validation accuracy
exceeding 99%
and final accuracy vs model size for each task.",
"Interestingness": 8,
"Feasibility": 7,
"Novelty": 7,
"novel": true
```

Idea 2/50 - optimizer_grokking

```
"Name": "optimizer_grokking",
"Title": "Optimization Dynamics and Grokking: Comparing SGD and Adam with
Different Learning Rate Schedules",
"Experiment": "Modify the training loop to support two optimizers (SGD,
Adam) and two learning rate schedules (constant, cosine annealing). For
each combination, run multiple experiments with different random seeds.
Track validation accuracy, training loss, and L2 norm of weight updates
throughout training. Compare the timing and extent of grokking across these
optimization strategies for each dataset. Analyze how different
optimization dynamics correlate with grokking behavior, including
statistical analysis of the results.",
"Interestingness": 9,
"Feasibility": 8,
"Novelty": 8,
"novel": true
```

Idea 3/50 - biased_data_grokking


```
"Name": "biased_data_grokking",
"Title": "Grokking Under Biased Data: The Effect of Input Range Bias on
Neural Network Generalization",
"Experiment": "Modify the fetch_train_example method in AbstractDataset to
introduce a simple bias: favoring lower-valued inputs. For modular
arithmetic operations, sample 70%
of the input range. For permutations, favor permutations with more elements
in their original positions. Keep the validation set unbiased. Run
experiments comparing grokking behavior on biased vs. unbiased training
sets. Track metrics such as steps to 99%
validation accuracy, and training loss. Analyze how this bias affects
grokking across different operations.",
"Interestingness": 8,
"Feasibility": 8,
"Novelty": 8,
"novel": true
```


Idea 4/50 - adaptive_noise_grokking


```
"Name": "adaptive_noise_grokking",
"Title": "Adaptive Noise in Grokking: Investigating Input Perturbations on
Algorithmic Learning and Representations",
"Experiment": "Modify the GroupDataset class to add operation-specific
noise during training: (1) For modular arithmetic, add small integer
perturbations. (2) For permutations, occasionally swap two elements.
Implement three noise levels (low, medium, high) for each operation.
Compare grokking behavior across noise levels and operations, tracking
steps to 99%
loss. Analyze learned representations by visualizing attention patterns and
performing principal component analysis (PCA) on hidden states at different
training stages. Compare these representations between noisy and non-noisy
training to understand how noise affects the abstraction of concepts.",
"Interestingness": 9,
"Feasibility": 8,
"Novelty": 9,
"novel": true
```


Idea 5/50 - attention_evolution_grokking


```
"Name": "attention_evolution_grokking",
"Title": "Attention Evolution in Grokking: Quantifying the Transition from
Memorization to Generalization",
"Experiment": "Modify the Transformer class to output attention weights.
Extract and store attention weights at key checkpoints: start, mid-
training, grokking point (99%
visualization tools for attention heatmaps and create plots showing
attention evolution over time. Calculate the Frobenius norm of the
difference between attention matrices at consecutive checkpoints to
quantify attention evolution. Compare attention patterns and evolution
metrics across different operations (modular arithmetic vs. permutations).
Analyze attention for specific, informative input sequences to enhance
interpretability. Correlate attention evolution metrics with validation
```